# SiamS$^2$F: Satellite Video Single-Object Tracking Based on a Siamese Spectral–Spatial-Frame Correlation Network

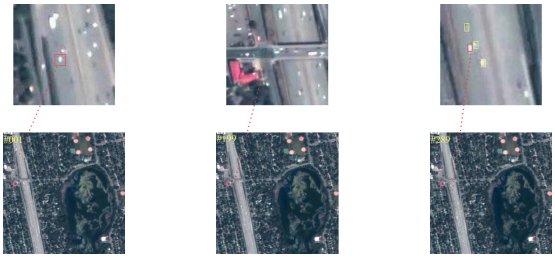
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***Abstract*— In recent years, satellite video object tracking has received widespread attention as a research hotspot, but is also faced with many challenges. In satellite video, moving objects usually consist of only a few pixels, which makes it more difficult for the tracker to distinguish the target from the background. Furthermore, occlusion factors such as clouds, trees, and bridges can bring challenges when relocating the target from adjacent frames. In this article, to solve the above-mentioned problems, we propose a Siamese spectral–spatial-frame (SiamS$^2$F) correlation network, which combines deep spectral, spatial, and frame features to enhance the interaction between video frames, to better focus on the target continuity. First, the deep features of the video are extracted with a Siamese backbone, along with a channel–spatial attention module. Second, a spectral–spatial-frame (SSF) module is proposed, where the output features of the backbone are fused based on the graph attention layer, and then the intraframe and interframe information of the fused features is enhanced by a newly designed multiframe interactive attention (MFIA) mechanism. Third, to solve the problem of similar objects, an additional center loss function is proposed in the classification regression head (CRH), where the search area is limited to a local range by adding an inbox identifier, to reduce the impact of the similar objects around the target. The proposed method was tested on two challenging benchmark remote sensing datasets—SatSOT and SV248S—where SiamS$^2$F outperformed the related state-of-the-art trackers.**

***Index Terms*— Object tracking, satellite video, Siamese network, spectral–spatial-frame (SSF) information.**

## I. Introduction

IN REMOTE sensing, satellite video observation has made great progress [1]. With the successful launch of video satellites such as SkySat-1, the Interface Region Imaging Spectrograph (IRIS), Jilin-1, and so on, it is now possible to monitor the motion information of ground objects [2], [3], [4] over a wide area. Many fields can be supported by remote sensing satellite video object tracking technology, such as traffic monitoring [5] and dynamic response to natural disasters [6], [7], [8], [9]. Although satellite video tracking technology has made some progress, it is still faced with some difficulties and challenges due to the long observation distance and complex imaging environment. The difficulties of satellite video target tracking are mainly concentrated in the following aspects, as shown in Fig. 1.



(a) (b) (c)

Fig. 1. Challenges of object tracking with satellite video. (a) Small target size. (b) Complex occlusions. (c) Similar objects.

1) *Small Target Size:* Compared with natural scenes, the moving targets in remote sensing videos only occupy a few pixels, as shown in Fig. 1(a), and it is difficult for the tracker to obtain enough spatial and shape information from the data.
2) *Complex Occlusions:* In satellite video, tracking can be interrupted by the occlusion problem. As shown in Fig. 1(b), it is very challenging for some trackers to retrack a moving target after losing its information.
3) *Similar Objects:* There can sometimes be some similar objects around the moving target, and these similar objects can bring difficulties in tracking the same target continuously. As shown in Fig. 1(c), when the moving target moves to the traffic jam area, it is difficult for some of the trackers to distinguish the target simply by a single-frame image.

To solve the above-mentioned problems, a number of satellite video tracking methods have been proposed. The correlation filter is a representative technique used to consider the motion information. Chen et al. [10] designed a rotation adaptive tracker with motion constraint (RAMC) to solve the object rotation and tracking drift problems. Li et al. [11] proposed the correlation particle Kalman filter (CPKF) to solve the problems of occlusion and rotation. Some deep

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learning-based methods have also been proposed. For example, Song et al. [12] proposed a new Siamese attention tracker to solve the problem of weak features and background noise in satellite target tracking. Yang et al. [13] proposed a Siamese tracking network (SiamMDM) to solve problems such as occlusions and small targets.

Although the existing satellite video tracking methods have achieved remarkable results in some specific scenarios with challenging attributes, the performance of the trackers could be further improved in complex situations containing comprehensive interference factors such as very small targets, similarity interference, and occlusions. These comprehensive interference factors not only increase the difficulty of the target tracking but also put forward higher requirements for the stability and accuracy of the algorithm. In addition to the above challenges, the existing target tracking algorithms are generally insufficient for extracting the intraframe and interframe information. Intraframe information mainly refers to the feature information of the target in the current frame, while interframe information involves the motion information and position relationship of the target in different frames, which is helpful for accurately locating the target in a complex scene. Therefore, in this article, considering the above-mentioned problems, a satellite video tracking method called the Siamese spectral–spatial-frame (SiamS$^2$F) correlation network is proposed. The major contributions of this work are as follows.

1) A novel Siamese tracker is proposed, which improves the robustness of the network with regard to similar objects and occlusion problems in complex scenes. The convolutional neural network (CNN)-based Siamese network is combined with a self-attention mechanism to take into consideration the global contextual information of the time series. The prior position information of the true bounding box and the predicted box is also considered in the loss function, to constrain the search area in a local range and reduce the similar objects.
2) To solve the problem of the small target size and occlusions, a spectral–spatial-frame (SSF) feature fusion module is proposed, which combines the spectral and spatial features to enhance the interaction between video frames, to better pay attention to the continuity of the target. In the SSF module, the channel attention and spatial attention features are fused based on a graph attention layer with a pixel-wise operation. The intraframe and interframe information of the fused features is then enhanced by the proposed multiframe interactive attention (MFIA) mechanism.
3) To solve the problem of similar objects, a classification regression head (CRH) is proposed, which consists of three loss functions. The cross-entropy (CE) loss function is improved to consider the imbalanced target and background. An additional center loss function is proposed to better locate the real target while reducing the impact of similar objects around it. To acquire the accurate location of the target, an improved distance intersection-over-union (DIoU) loss function is proposed by considering the distance between the true bounding box and the predicted box.

## II. Related Work

### A. Correlation Filter-Based Tracking Methods

The correlation filter was first applied in the field of Fourier signal processing [14] and has since been studied in target tracking. The correlation of the candidate target region and the filter template is computed, and the target position can be located by the maximum response value of the position. The minimum output sum of squared error (MOSSE) algorithm was proposed by Bolme et al. [15] and was the first attempt at correlation filter tracking. The kernelized correlation filter (KCF) [16] considers the problem of fast and accurate target tracking in a real-time video stream by using the filter and cyclic matrix to efficiently process the target features in the frequency domain.

In recent years, a series of correlation filter-based target tracking methods have also emerged in the remote sensing field. For example, Du et al. [17] designed a satellite video target tracking algorithm that integrates a KCF tracker and a three-frame differential algorithm to effectively distinguish between target and background information. Du et al. [3] proposed a multiframe optical flow tracker, which can achieve a better tracking effect while detecting slight target motion. Shao et al. [18] combined the velocity characteristics with a filter to solve the model drift problem. Guo et al. [19] effectively solved the problem of tracking failure by introducing high-speed correlation filtering. Wang et al. [20] solved the problem of weak target features by considering a sample training strategy and sample characterization to enhance satellite video target tracking. Xuan et al. [21] proposed that the CFME algorithm can effectively suppress the boundary effect and occlusion problems. Li et al. [11] proposed the CPKF framework to solve the problems of target occlusion and background clutter.

### B. Deep Learning-Based Tracking Methods

Deep neural networks can learn feature representations and motion patterns of the target from large amounts of training data, enabling a more accurate and robust performance. The current deep learning-based trackers can be divided into four main categories: anchor-based trackers, anchor-free trackers, hybrid correlation filter and neural network-based trackers, and transformer-based trackers. The anchor-based trackers generate proposal regions to enhance the tracking accuracy. For example, SiamRPN [22] introduces the region proposal network (RPN) to SiamFC by generating candidate regions to further improve the accuracy of the target location. SiamRPN++ [23] introduces a spatially aware sampling strategy to solve the problem of the insufficient translation invariance of the traditional Siamese trackers. SiamFC [24] is an anchor-free tracker, which uses an end-to-end Siamese architecture to determine the location of the target by directly comparing the feature similarity between the target and candidate regions, achieving an efficient tracking performance. SiamBAN [25] uses a fully convolutional network (FCN) to solve the cumbersome configuration problems of multiscale search and predefined anchor boxes existing in the anchor-based trackers. SiamGAT [26] is a twin network with a

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graph attention mechanism (GAM) to better express the target features. The filter-based methods can also be combined with a neural network to enhance the time-series information. The accurate tracking by overlap maximization (ATOM) method [27] combines filter-based tracking with deep features to estimate the target state and the change of the target more effectively. DiMP [28] is a new end-to-end tracking framework aimed at solving the problem of inaccurate tracking under complex scenes. PrDiMP [29] is a probabilistic regression method, which effectively solves the uncertainty problem of position prediction in target tracking. KeepTrack [30] introduces a training strategy that combines partial annotation and self-supervision to solve the problem of the fuzzy relationship between the context and target in tracking. With the development of transformer models [31], these models have also been studied for tracking tasks. The TransT method [32] uses a transformer structure and a mixed attention mechanism to consider the influence of occlusions. TrDiMP [33] extracts and fuses the target features by using self-attention and cross-attention mechanisms. ToMP [34] uses the predictor of the transformer model for the tracking, to improve the insufficient expressive ability of the tracking network. AiATrack [35] uses an attention-in-attention (AiA) module to solve the noise and blur problem, which can lead to inaccurate tracking. SeqTrack [36] is a sequence-to-sequence tracking framework, where a simple encoder–decoder transformer structure is used to simplify the complicated head network and loss function. LoReTrack [37] introduces a low-resolution transformer tracking algorithm to solve the video degradation problem. VastTrack [38] utilizes a dataset covering rich attribute categories for better testing of the transformer-based methods. However, satellite video tracking encounters the problem of a complex background and weak target, and the performance of the tracking networks in natural scenes needs to be further improved for use in satellite video.

In recent years, a series of deep learning-based target-tracking methods have also been proposed in the field of satellite video tracking. For example, Shao et al. [2] proposed a lightweight parallel network for locating objects consisting of only a few pixels. Cui et al. [39] proposed a target tracking method based on deep reinforcement learning (DRL) for the situation of complete target occlusion. The two above-mentioned tracking methods incorporate a re-detection mechanism, which considers the information of the adjacent frames when the target in the current frame cannot be located. Zhang et al. [4] designed a target tracking framework that tracks by learning information about the satellite video vehicle movement and context. Yang et al. [13] proposed a SiamMDM, which fuses multiple response maps and introduces spatiotemporal constraints to solve problems such as occlusions and small objects. However, the above-mentioned methods all have limitations in considering the multiple challenges existing in a scene, and the long-range contextual information of the time series is not well explored.

## III. Proposed SiamS$^2$F Framework

In this section, we present a detailed description of the proposed SiamS$^2$F framework. As shown in Fig. 2, the network structure of SiamS$^2$F is composed of three main parts.

1) Siamese backbone. GoogLeNet [40] is adopted in the backbone to learn multiscale feature representations with fewer parameters. A channel and spatial attention module (CSM) was then followed to better extract the deep features.
2) SSF attention module. The SSF module consists of a GAM and an MFIA mechanism. The former can achieve point-to-point information embedding matching and can adaptively consider the size and rotation changes of different objects. The latter can enhance the information within the video frame and the dependencies between frames to achieve higher-quality pixel regression.
3) Bounding box prediction head. The CRH is selected as the basic structure, which consists of two loss functions—classification loss and regression loss—and this is extended with an additional center loss function. To solve the problem of foreground and background imbalance, the classification and regression loss functions are designed to suppress the complex background. The center loss function is designed to better distinguish the real target from similar objects.

### A. Siamese Backbone

A Siamese structure with shared parameters is beneficial for finding the true target from the search areas and improving the network efficiency. The Siamese backbone has two inputs—a template branch with tracking template $z$ as input and a search branch with search area $x$ as input—and the corresponding response maps $F_z$ and $F_x$ are acquired. Due to the weak spectral and shape characteristics of the targets in satellite video, the CSM is added to initially enhance the spectral and spatial features. The response maps are input into the CSM to obtain the output features $F_{\mathrm{t}}$ and $F_s$ for the template branch and search branch, respectively.

The operation in the CSM can be denoted as

$$F' = F \otimes (R_c(F) \otimes R_s(F)) \tag{1}$$

where $\otimes$ denotes the element-wise product. $F$ stands for $F_z$ and $F_x$, and $F'$ stands for $F_{\mathrm{t}}$ and $F_s$. $R_c(\cdot)$ is the channel attention and $R_s(\cdot)$ is the spatial attention

$$R_{\mathrm{C}}(F) = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F))) \tag{2}$$

$$R_{\mathrm{S}}(F) = \sigma(\mathrm{Conv2d}(\mathrm{AvgPool}(F)||\mathrm{MaxPool}(F))) \tag{3}$$

where $\sigma$ denotes the sigmoid function, $\|$ is the operation of concatenating features, and Conv2d denotes the 2-D convolution operation.

### B. SSF Attention

The output feature $F'$ from the backbone is then input into the SSF module. The SSF module is designed mainly to solve the limitations of the traditional Siamese framework in long-sequence modeling. Strengthening the global interframe information is beneficial for tracking a target with weak spectral and shape information in satellite video data. The

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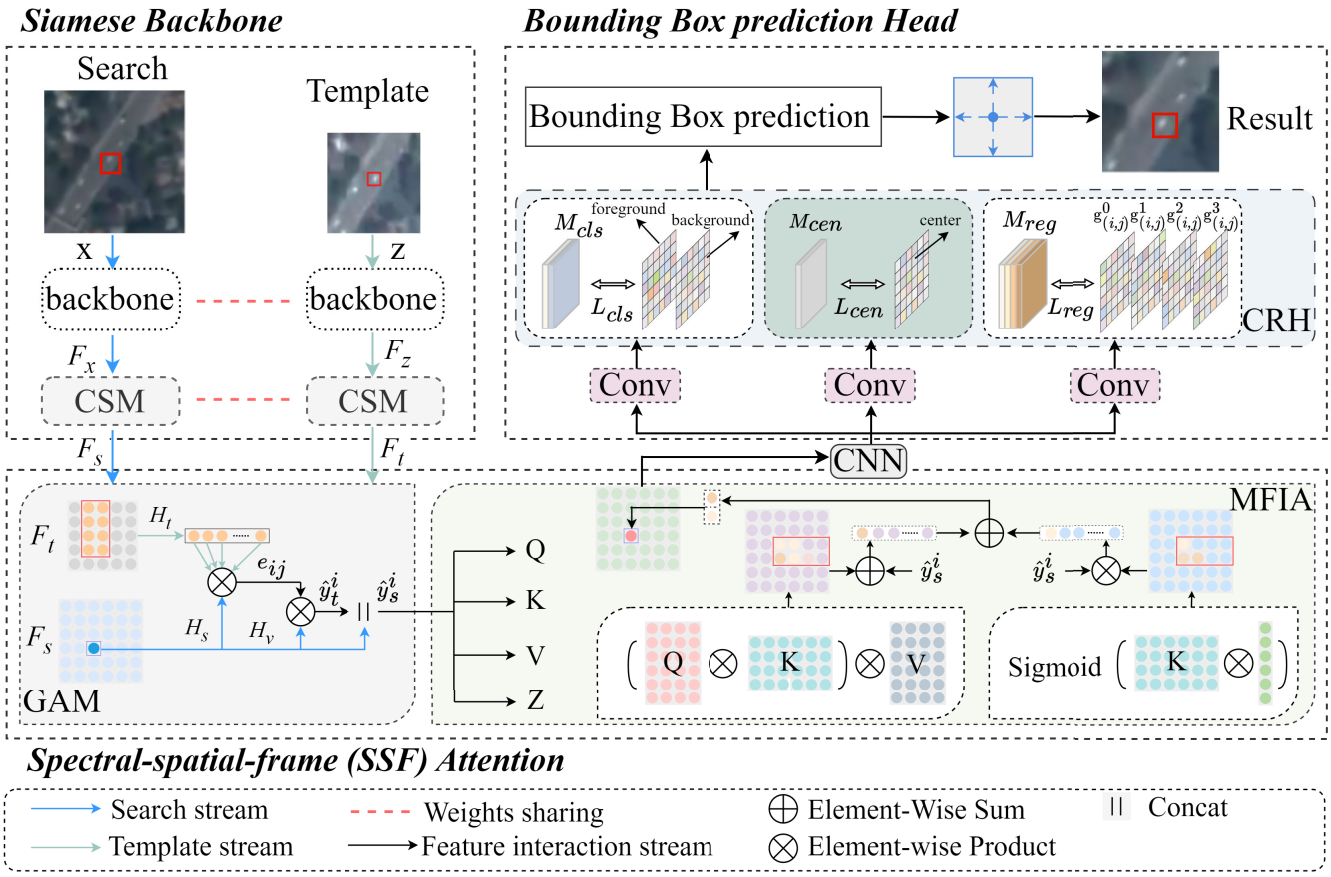



Fig. 2. Network structure of SiamS$^2$F is composed of three main modules: on the left is the Siamese backbone for feature extraction, and below this is the SSF feature fusion module, which combines the spectral and spatial features to enhance the interaction between video frames, to better focus on the continuity of the target. On the right is the CRH, which consists of two loss functions—classification loss and regression loss—and is extended with an extra center loss function.

SSF module contains two main parts: the GAM and the MFIA module. The GAM is used to generate spectrally and spatially invariant features and the MFIA module is designed to consider the correlation of the tracked target in multiple frames.

*1) Graph Attention Mechanism (GAM) :* Most of the existing tracking methods are based on matching the target and the search area. However, these methods have difficulty in considering the problem of the scale and rotation change of the target between frames. To solve this problem, the GAM [41] is used, which represents the target feature and template feature as a graph, to compute the correlation of the local feature point in the target area with the global points in the search area and reduce the impact of the rotation and scale change.

We let $V_t$ and $V_s$ be the node sets for $F_t$ and $F_s$, respectively. $E_{ij}$ denotes the cross correlation score of nodes $i$ and $j$. $E_{ij}$ is calculated as follows:

$$E_{ij} = \left(H_s \mathrm{y}_s^i\right)^T \left(H_t \mathrm{y}_t^j\right) \tag{4}$$

where $\mathrm{y}_s^i$ is the feature vector of node $i$ and $\mathrm{y}_t^j$ is the feature vector of $j$. $H_s$ and $H_t$ are the linear transformation matrices for the target feature and template feature, respectively.

A softmax function is used to normalize the cross correlation score $E_{ij}$

$$e_{ij} = \frac{\exp\left(E_{ij}\right)}{\sum_{j \in V_t} \exp\left(E_{ij}\right)} \tag{5}$$

where $e_{ij}$ is the generated score after normalization.

The weighted feature of node $i$ after the graph attention operation can be acquired by the following equation:

$$\hat{\mathrm{y}}_t^i = \sum_{j \in V_t} e_{ij} H_v \mathrm{y}_t^j \tag{6}$$

where $\hat{\mathrm{y}}_t^i$ is the feature after the graph attention operation and $H_v$ is the linear transformation matrix.

Finally, the weighted feature is fused with the node feature $\mathrm{y}_s^i$ to obtain a feature representation $\hat{\mathrm{y}}_s^i$

$$\hat{\mathrm{y}}_s^i = \mathrm{ReLu}\left(\hat{\mathrm{y}}_t^i \parallel \left(H_v \mathrm{y}_s^i\right)\right) \tag{7}$$

where $\parallel$ is the concat operation.

*2) MFIA Mechanism :* The MFIA module takes into account the interframe information in the satellite video data. In the MFIA module, to consider the global interframe information for each batch, a self-attention mechanism is utilized, which introduces the interaction of three matrices $Q$, $K$, and $V$ to enhance the key information in the video from the aspect

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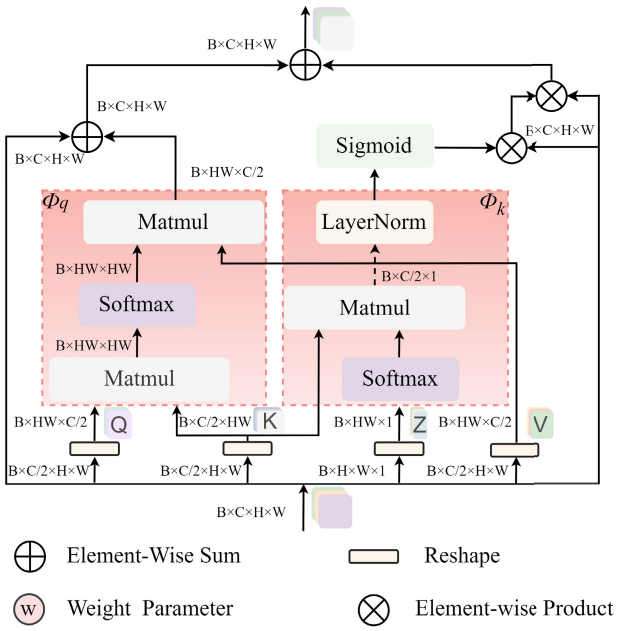



Fig. 3. Architecture of the MFIA module.

of the time series. Then, due to the weak shape information of the target in satellite video, an extra $Z$ matrix is added to the MFIA module to enhance the spatial information in each frame, through the interaction between the $K$ matrix and the $Z$ matrix. The output of the MFIA module can be acquired based on (9)–(11), where the $Z$ matrix has the dimension of B × HW × 1 and is used to express the weight of each pixel in the spatial domain in each frame. Through the MFIA module, the feature transfer between the template branch and the search branch can be improved, and the small changes in the video frame and the long-distance dependency between frames can be obtained, to achieve higher-quality pixel regression. Even after the target is occluded, good tracking can be achieved.

The overall structure of the MFIA module is shown in Fig. 3. Given an intermediate feature map $\hat{y}_s^i$ generated by the GAM as the input, a nonlinear projection of $\hat{y}_s^i$ based on convolution and reshape operations is performed and four matrices $Q$, $K$, $V$, and $Z$ are obtained

$$Q, K, V, Z = \text{reshape}\left(\text{Conv2d}\left(\hat{y}_s^i\right)\right) \tag{8}$$

where the dimensions of $Q$, $K$, $V$, and $Z$ are B × HW × C/2, B × C/2 × HW, B × HW × C/2, and B × HW ×1, respectively.

The attention weight is then acquired by matrix multiplication and the softmax operation

$$\Phi_{qk} = \frac{\exp\left(QK^T\right)}{\sum_{i,j=1}^{N}\exp\left(QK^T\right)} \tag{9}$$

where $N$ is the dimension of $QK^T$.

The feature information in $V$ and $\Phi_{qk}$ is then used to perform a matrix multiplication operation and further aggregate the feature information to obtain $\Phi_q$

$$\Phi_q = \text{Conv2d}\left(\Phi_{qk}V\right) \tag{10}$$

At the same time, the channel information in $\hat{y}_s^i$ is extracted to obtain $Z$, and matrix multiplication operations are performed on $K$ and $Z$ to enhance the spatial weight of the key points related to the target and acquire $\Phi_k$

$$\Phi_k = \text{Sigmoid}\left(L_{\text{LN}}\text{Conv2d}\left(K\frac{\exp(Z)}{\sum_{i=1}^{N}\exp(Z)}\right)\right) \tag{11}$$

The overall attention process can be summarized as

$$\text{MFIA} = \left(\mu\Phi_q + \hat{y}_s^i\right) + \left(\left(\upsilon\left(\Phi_k \otimes \hat{y}_s^i\right)\right) \otimes \hat{y}_s^i\right) \tag{12}$$

where Conv2d is the 1 × 1 convolution layer, $L_{\text{LN}}(\cdot)$ is the LayerNorm operator, and $\mu$ and $\upsilon$ represent the weight parameters.

### C. Bounding Box Prediction Head

As shown in Fig. 2, the bounding box prediction head consists of the classification module and the regression module applied in a pixel-by-pixel manner. The classification module consists of two parts: a classification branch and a center branch. The classification loss $L_{\text{cls}}$ focuses on distinguishing the background and target information, while the regression loss $L_{\text{reg}}$ helps to determine the accurate position of the target. However, the two current types of loss functions still have some limitations in distinguishing between a real target and similar objects around it. Therefore, the center loss $L_{\text{cen}}$ is proposed, which improves the DIoU loss by adding an inbox identifier to determine whether the point in the patch to be searched falls within the bounding box delineated in the first frame.

The classification feature map $M_{\text{cls}}$, the center feature map $M_{\text{cen}}$, and the regression feature map $M_{\text{reg}}$ are obtained from the output of the SSF module based on the convolutional layers. The overall loss is calculated as follows:

$$L = \alpha L_{\text{cls}} + \beta L_{\text{cen}} + \gamma L_{\text{reg}} \tag{13}$$

where $L_{\text{cls}}$ represents the improved CE loss, $L_{\text{reg}}$ represents the improved DIoU loss, and $L_{\text{cen}}$ represents the center loss. $\alpha$, $\beta$, and $\gamma$ in the above formula represent the weight parameters.

The CE loss can be calculated based on the following equations:

$$L_{\text{cls}} = \frac{-1}{N}\sum_{I==1}\left(T(i,j)\log M_{cls}(i,j) + (1-T(i,j))\log(1-M_{cls}(i,j))\right) \tag{14}$$

$$T(i,j) = \frac{\min\left(g_{(i,j)}^0, g_{(i,j)}^2\right)\min\left(g_{(i,j)}^1, g_{(i,j)}^3\right)}{\max\left(g_{(i,j)}^0, g_{(i,j)}^2\right)\max\left(g_{(i,j)}^1, g_{(i,j)}^3\right)}I \tag{15}$$

$$I = \begin{cases} 1 & \text{if } g_{(i,j)}^v > 0, v = 0,1,2,3 \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

where $T(i,j)$ represents the distance between feature point $(i,j)$ in feature map $M_{\text{cls}}$ to the bounding box in the first frame and $N$ is the number of feature points in $M_{\text{cls}}$. $g_{(i,j)}^0$ $g_{(i,j)}^1$ $g_{(i,j)}^2$ and $g_{(i,j)}^3$ are the distances between $(i,j)$ and the four coordinates of the bounding box. The incorporation of the inbox identifier $I$ is aimed at determining whether the point in the patch to be searched falls within the bounding

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TABLE I
DESCRIPTIONS OF THE TEN ATTRIBUTES AND DEFINITIONS IN THE SV248S DATASET

| Attribute | Definition |
|---|---|
| STO | **Short-Term Occlusion:The sequence exists less than or equal to 50 consecutive frames with occlision flags** |
| LTO | **Long-Term Occlusion:The sequence exists more than 50 consecutive frames with occlusion flags** |
| DS | **Dense Similarity:One or more similar objects exist around the tracked object in the range of 2.5 times object size** |
| IV | **Illumination Variation:The object has noticeable changes in brightness or color** |
| BCH | **Background Change:The background of the tracked object has noticeable changes in color or texture** |
| SM | **Slow Motion:The moving speed of the tracked object is less than 2.2 pixels per second** |
| ND | **Natural Disturbance:The moving speed of the tracked object is less than 2.2 pixels per second** |
| CO | **Natural Disturbance:STO or LTO occur two or more times in a sequence** |
| BCL | **Background Cluster:There are at least ten frames that contain the invisible flag** |
| IPR | **In-Plane Rotation:The object has an in-plane rotation at an angle greater than or equal to 30 degrees** |

box delineated in the first frame. Therefore, the search area can be limited to within the local region, to reduce the impact of similar objects around the target. $I = 1$ indicates that point $(i, j)$ falls in the range of the bounding box of the first frame. The CE loss function aims to focus on the points around the range of the box in the first frame in the classification, to reduce the impact of background confusion.

To consider the impact of similar objects around the true target, an extra center loss function is designed based on the original CE loss

$$L_{\text{cen}} = \frac{-1}{N}\sum_{I==1}\Gamma_{\omega}\left(1-\Gamma_{M_{\text{cen}}(i,j)}\right)^{\psi}\left(T(i,j)\log M_{\text{cen}}(i,j) + (1-T(i,j))\log(1-M_{\text{cen}}(i,j))\right) \tag{17}$$

$$\Gamma_{\omega} = \omega T(i,j) + (1-\omega)(1-T(i,j)) \tag{18}$$

$$\Gamma_{M_{\text{cen}}(i,j)} = (1-T(i,j))(1-M_{\text{cen}}(i,j)) + T(i,j)M_{\text{cen}}(i,j) \tag{19}$$

where $\Gamma_{\omega}$ is a coefficient related to $T(i, j)$ to suppress the weight of the easy candidate points and $\omega \in (0, 1)$. $(1 - \Gamma_{M_{\text{cen}}(i,j)})^{\psi}$ is the coefficient related to the feature vector $M_{\text{cen}}$ of the search patch, which also plays the role of reducing the weight of the easy candidate points and increasing the weight of the hard candidate points. When a candidate point is away from the bounding box center, the value of $T(i, j)$ decreases, and the weight of the loss of this point increases.

For the regression task, an improved DIoU [42] loss function is proposed

$$L_{\text{reg}} = \frac{1}{N}\sum_{i,j}\left(1 - L_{\text{IoU}}\left(M_{\text{reg}}(i,j)\right) + \frac{D^2(h,h_{\text{t}})}{b^2}\right)I \tag{20}$$

where $h$ and $h_{\text{t}}$ are the center points of the prediction box and the true target box, respectively. $D(\cdot)$ is the Euclidean distance and $b$ is the diagonal length of the minimum bounding box that encloses the prediction and true target boxes. $L_{\text{IoU}}$ is the intersection over union (IoU) [43] loss function.

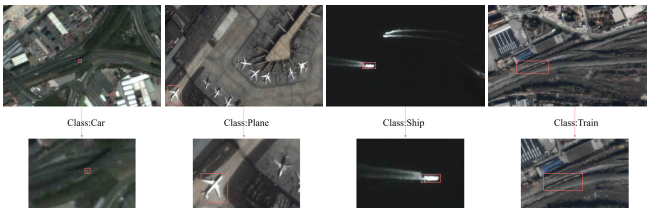



Fig. 4. Four classes in the SatSOT dataset.




Fig. 5. Four classes in the SV248S dataset.

## IV. EXPERIMENTS

### A. Experimental Setup

*1) Dataset:* The proposed method was trained on five datasets: ImageNet VID [44], YouTube-BB [45], COCO [46], ImageNet DET [44], and GOT10K [47]. Two satellite video datasets—SatSOT [48] and SV248S [49]—were then used to comprehensively evaluate the proposed method (a detailed description of these attributes can be found in Table I). As shown in Fig. 4, the SatSOT dataset contains four categories—trains, planes, cars, and ships—with an average bounding box size of 395, 566.3, 112.4, 1714.3, and 48.5 pixels, respectively. More than 70% of the video sequences have a bounding box size smaller than 1000 pixels. The dataset consists of 105 video sequences with 11 interference attributes, such as occlusion and illumination, ranging in length from 120 to 750 frames, for a total of 27 664 frames.

There are also four categories of objects in the SV248S dataset, as shown in Fig. 5, that is, vehicles, large vehicles, ships, and airplanes. The dataset consists of 248 video sequences, each with a length range of 500 to 753 frames.

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This dataset mainly focuses on small objects and brings more challenges.

*2) Evaluation Metrics:* To evaluate the performance of the trackers, the widely used object tracking benchmark (OTB)-2015 [50] paradigm is followed, which uses the one-pass evaluation (OPE) method [51] with precision and success plots [52], [53]. These plots are based on two evaluation metrics: the center location error (CLE) and the IoU.

The CLE is calculated based on the following equation:

$$\mathrm{CLE} = \sqrt{(a_1 - a_2)^2 + (b_1 - b_2)^2} \tag{21}$$

when the Euclidean distance between the center of the predicted bounding box $(a_1, b_1)$ and the center of the real bounding box $(a_2, b_2)$ is less than the given threshold, the tracking is considered successful. The precision plot shows the percentage of frames in the video sequence where the CLE is less than the threshold. In the experiments conducted in this study, the threshold was set to 5 according to [13].

The IoU is calculated based on the following equation:

$$\mathrm{IoU} = \frac{|r_t \cap r_a|}{|r_t \cup r_a|} \tag{22}$$

where $r_t$ is the predicted bounding box and $r_a$ is the true target box. The success plot shows the relative number of frames where the IoU exceeds the threshold, where the general threshold is set to 0.5. We use the precision and success plots to rank all the trackers.

The frames per second (FPS) are used to evaluate the speed of the trackers.

*3) Implementation Details:* The platform used in the experiments was a 64-bit Ubuntu 20.04 workstation with a 24-GB GeForce RTX 3090 GPU. The experimental environment comprised CUDA 11.0, Python 3.7, and PyTorch 1.7.1. The proposed method was trained for 30 epochs. The learning rate was set to 0.005 for the first five epochs, and then linearly increased to 0.01, decaying to 0.0005 for the last 25 epochs. The batch size was set to 256. The size of the initial frame template image was set to 127 × 127 pixels, and the size of the search area image was 287 × 287 pixels.

## B. Ablation Study

The ablation study was performed on the SatSOT and SV248S datasets to validate each proposed module in the SiamS$^2$F framework. The ablation experiments were divided into two parts. In the first part, we analyzed the CSM, SSF, and CRH modules, and in the second part, we analyzed the GAM and MFIA modules contained in the SSF module by fixing the CSM and CRH modules. The visual results obtained by adding each module individually to the baseline [24] are also provided in Fig. 6, to further analyze the performance of the modules.

As shown in Table II, on the SatSOT dataset, when only one module is used, the SSF module acquires the highest accuracy, which indicates that considering the SSF information is beneficial for the satellite video tracking task. By integrating the interaction between frames, the motion information helps the tracker better locate the small targets. The combination

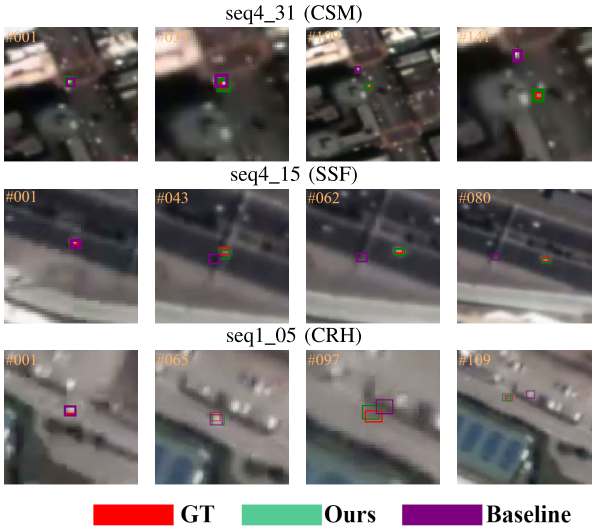



Fig. 6. Examples of the effect of adding the different modules to the baseline method.

of two modules is better than using a single module, and the best combination is SSF + CRH, which achieves an accuracy of 61.2% and 48.0% in precision and success, respectively. All three modules work best together, with the precision and success reaching 62.1% and 48.6%, respectively, which are 0.9% and 0.6% higher than the two-module combination. However, in terms of the running time, the running time does increase as the module number increases, and using all three modules takes the most time for training, but the difference is not very large. The results for the SV248S dataset are similar to those for the SatSOT dataset, and the combination of two modules is again better than the use of a single module, in which the SSF + CRH combination is superior, with the precision and success reaching 74.1% and 49.3%, respectively. Using all three modules works best of all, with the precision and success reaching 76.3% and 50.6%, respectively, which are 2.2% and 1.3% higher than the best combination of two modules.

To make a further ablation analysis of the proposed modules, a visual analysis has also been provided for the case of each module working independently. As shown in Fig. 6, the seq4_31 sequence contains four challenge attributes: STO, DS, IV, and IPR. When frame 28 is reached, the baseline method loses the target because it is difficult to capture enough information, due to the similarity of the target to the background information. In contrast, when the CSM is added, it ensures continuous and accurate tracking due to its spectral and spatial feature extraction capabilities. There is an occlusion problem in the seq4_15 sequence, and when the target passes the bridge after frame 43, the baseline method fails to track, but the global SSF module successfully recaptures and stabilizes the target after the occlusion by strengthening the intraframe information integration and interframe dependencies. The main problem of the seq1_05 sequence is the similarity of the surrounding objects. From frame 35, many similar objects appear around the target, which causes the failure of the baseline method. When the CRH is used, correct tracking is achieved to 230 frames because the search box is adaptively located by the center loss.

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TABLE II
ABLATION STUDY FOR THE DIFFERENT COMPONENTS OF THE SIAMS$^2$F FRAMEWORK ON THE SV248S AND SATSOT DATASETS

| CSM | SSF | CRH | SatSOT | | Sv248S | | Speed |
|---|---|---|---|---|---|---|---|
| | | | Precision(%) | Success(%) | Precision(%) | Success(%) | (FPS) |
| × | × | × | 46.7 | 35.0 | 67.2 | 39.0 | 93.3 |
| ✓ | × | × | 49.0(↑ 2.3) | 37.9(↑ 2.9) | 68.5(↑ 1.3) | 41.3(↑ 2.3) | 84.8 |
| × | ✓ | × | 50.1(↑ 3.4) | 38.7(↑ 3.7) | 71.8(↑ 4.6) | 41.7(↑ 2.7) | 84.2 |
| × | × | ✓ | 53.2(↑ 6.5) | 40.4(↑ 5.4) | 67.6(↑ 0.4) | 43.8(↑ 4.8) | 87.6 |
| ✓ | ✓ | × | 51.2(↑ 4.5) | 41.0(↑ 6.0) | 72.3(↑ 5.1) | 45.9(↑ 6.9) | 83.4 |
| ✓ | × | ✓ | 56.5(↑ 9.8) | 40.8(↑ 5.8) | 71.0(↑ 3.8) | 45.3(↑ 6.3) | 84.3 |
| × | ✓ | ✓ | 61.2(↑ 14.5) | 48.0(↑ 13.0) | 74.1(↑ 6.9) | 49.3(↑ 10.3) | 84.2 |
| ✓ | ✓ | ✓ | 62.1(↑ 15.4) | 48.6(↑ 13.6) | 76.3(↑ 9.1) | 50.6(↑ 11.6) | 82.2 |

TABLE III
ABLATION STUDY FOR THE DIFFERENT COMPONENTS OF THE SSF MODULE PERFORMED ON THE SV248S AND SATSOT DATASETS

| GAM | MFIA | SatSOT | | Sv248S | | Speed |
|---|---|---|---|---|---|---|
| | | Precision(%) | Success(%) | Precision(%) | Success(%) | (FPS) |
| × | × | 56.5 | 40.8 | 71.0 | 45.3 | 84.3 |
| × | ✓ | 57.6 | 43.0 | 71.9 | 45.7 | 83.6 |
| ✓ | × | 59.0 | 47.4 | 71.6 | 48.3 | 83.1 |
| ✓ | ✓ | 62.1 | 48.6 | 76.3 | 50.6 | 82.2 |

As shown in Table III, the performance of the GAM and MFIA modules contained in the SSF module was further analyzed by fixing the CSM and CRH modules. First, compared with not using these two modules, adding the GAM and MFIA modules improves the accuracy of the two datasets. With the use of the GAM module, the precision and success reach 59.0% and 47.4% on the SatSOT dataset, and 71.6% and 48.3% on the SV248S dataset, respectively. With the addition of the MFIA module, the precision and success reach 57.6% and 43.0% on the SatSOT dataset, and 71.9% and 45.7% on the SV248S dataset, respectively. The highest accuracy is achieved when the GAM and MFIA modules are both used. For the speed of the network, because the graph attention and self-attention mechanisms are used in the GAM and MFIA modules, the speed is slightly decreased when these modules are added.

### C. Comparison With State-of-the-Art Trackers

*1) Introduction of the Compared Trackers:* Fifteen representative trackers were evaluated on the SatSOT dataset and the SV248S dataset. The Python version of the OTB evaluation toolkit was used. The trackers included one correlation filter-based method (CFME [21]), nine CNN-based methods (ATOM [27], DiMP50 [28], PrDiMP50 [29], KeepTrack [30], SiamFC [24], SiamRPN [22], SiamBAN [25], SiamGAT [26], and SiamMDM [13]), and five Transformer-based methods (TransT [32], TrDiMP [33], ToMP50 [34], AiATrack [35], and SeqTrack [36]).

*2) Accuracy Analysis:* Table IV lists the accuracy of all the different algorithms on the two datasets. For the SatSOT dataset, the precision and success of the proposed SiamS$^2$F framework are 62.1% and 48.6%, respectively, which are 1.0% and 2.5% higher than the second-best SiamMDM algorithm. On the SV248S dataset, SiamS$^2$F achieves a precision of 76.3% and a success of 50.6%, which represents a significant improvement over the second-best method, of 4.7% and 5.5%, respectively. A comparison of the parameters and floating point operations (FLOPs) is also provided in Table IV. As shown in Table IV, for the FLOPs, it can be observed that the traditional CNN-based methods such as SiamFC (for which AlexNetV1 [54] was used as the backbone, and the scale_num was set to 3) and SiamRPN are significantly faster than the Transformer-based models, which can mainly be attributed to the convolution operations used in CNNs requiring less time cost. Despite utilizing the self-attention mechanism in the MFIA module, the proposed SiamS$^2$F framework can still maintain a reasonable efficiency level of 82.2 FPS because the shallow Siamese CNN structure is used to aggregate the spectral and spatial features, and the GAM is introduced to limit the search range. This shows that the proposed SiamS$^2$F framework can effectively combine the advantages of a CNN and a self-attention mechanism and has a moderate computational cost when compared to the transformer-based models. For the parameters, the performance is slightly better than the performance in FLOPs, reaching 14.43 M, which shows that the parameters of the proposed SiamS$^2$F framework are also at a moderate level.

Fig. 7 gives the precision plots and the success plots of the different methods. As shown in Fig. 7, the overall results of the different methods on the two datasets show that the proposed SiamS$^2$F framework achieves the best performance on both datasets. To analyze the performance of the tracking

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TABLE IV

OVERALL PERFORMANCE ON THE SATSOT AND SV248S DATASETS, WITH THE BEST THREE PERFORMANCES HIGHLIGHTED IN RED, BLUE, AND GREEN COLORS, RESPECTIVELY

| Tracker | Source | SatSOT | | SV248S | | Speed | FLOPs | Parameters |
|---|---|---|---|---|---|---|---|---|
| | | Precision(%) | Success(%) | Precision(%) | Success(%) | (FPS) | (G) | (M) |
| SiamS$^2$F | Ours | 62.1 | 48.6 | 76.3 | 50.6 | 82.2 | 19.67 | 14.43 |
| SiamMDM | TGRS2023 | 61.1 | 46.1 | 71.6 | 45.1 | 110.4 | 5.038 | 3.75 |
| SeqTrack | CVPR2023 | 51.2 | 40.8 | 37.2 | 16.1 | 22.20 | 66 | 89 |
| AiATrack | ECCV2022 | 39.2 | 32.9 | 28.1 | 13.9 | 23.34 | 9.45 | 17.95 |
| ToMP50 | CVPR2022 | 49.6 | 37.6 | 38.0 | 16.1 | 33.2 | 10.97 | 8.54 |
| SiamGAT | CVPR2021 | 56.7 | 43.9 | 69.5 | 45.3 | 90.2 | 17.28 | 14.23 |
| TrDiMP | CVPR2021 | 59.6 | 46.5 | 66.3 | 38.4 | 21.23 | 16.59 | 43.75 |
| TransT | CVPR2021 | 50.4 | 38.1 | 39.1 | 17.4 | 33.4 | 16.71 | 23.02 |
| KeepTrack | ICCV2021 | 46.3 | 38.3 | 49.8 | 31.2 | 20.1 | 28.65 | 43.10 |
| CFME | TGRS2020 | 55.0 | 41.5 | 42.4 | 27.5 | 266.63 | - | - |
| SiamBAN | CVPR2020 | 46.7 | 35.0 | 67.2 | 39.0 | 93.3 | 19.46 | 14.03 |
| PrDiMP50 | CVPR2020 | 49.6 | 40.8 | 49.9 | 32.0 | 22.91 | 15.51 | 43.1 |
| DiMP50 | ICCV2019 | 53.3 | 41.9 | 60.3 | 36.0 | 27.46 | 10.35 | 43.1 |
| ATOM | CVPR2019 | 53.7 | 41.7 | 62.7 | 36.3 | 31.28 | 10.97 | 8.54 |
| SiamRPN | CVPR2018 | 52.6 | 38.0 | 60.0 | 29.1 | 293.37 | 4.7 | 5.2 |
| SiamFC | ECCV2016 | 57.8 | 43.9 | 64.8 | 40.7 | 137.6 | 2.7 | 2.3 |

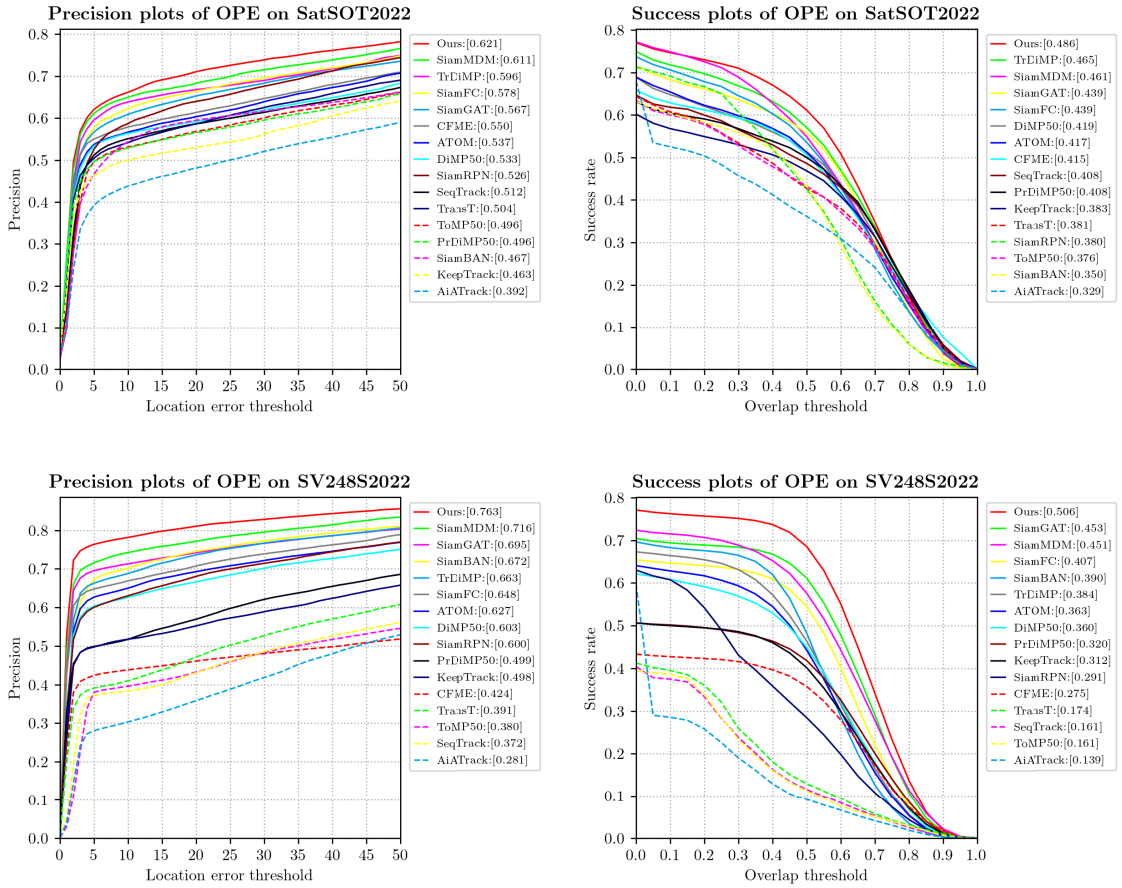



Fig. 7. Precision and success plots for the SatSOT and SV248S datasets.

algorithms under different unfavorable factors, the precision and success plots of each algorithm with different attributes of the SV248S dataset are displayed in Fig. 8. These attributes are BCH, BCL, CO, DS, IPR, IV, LTO, ND, SM, and STO.

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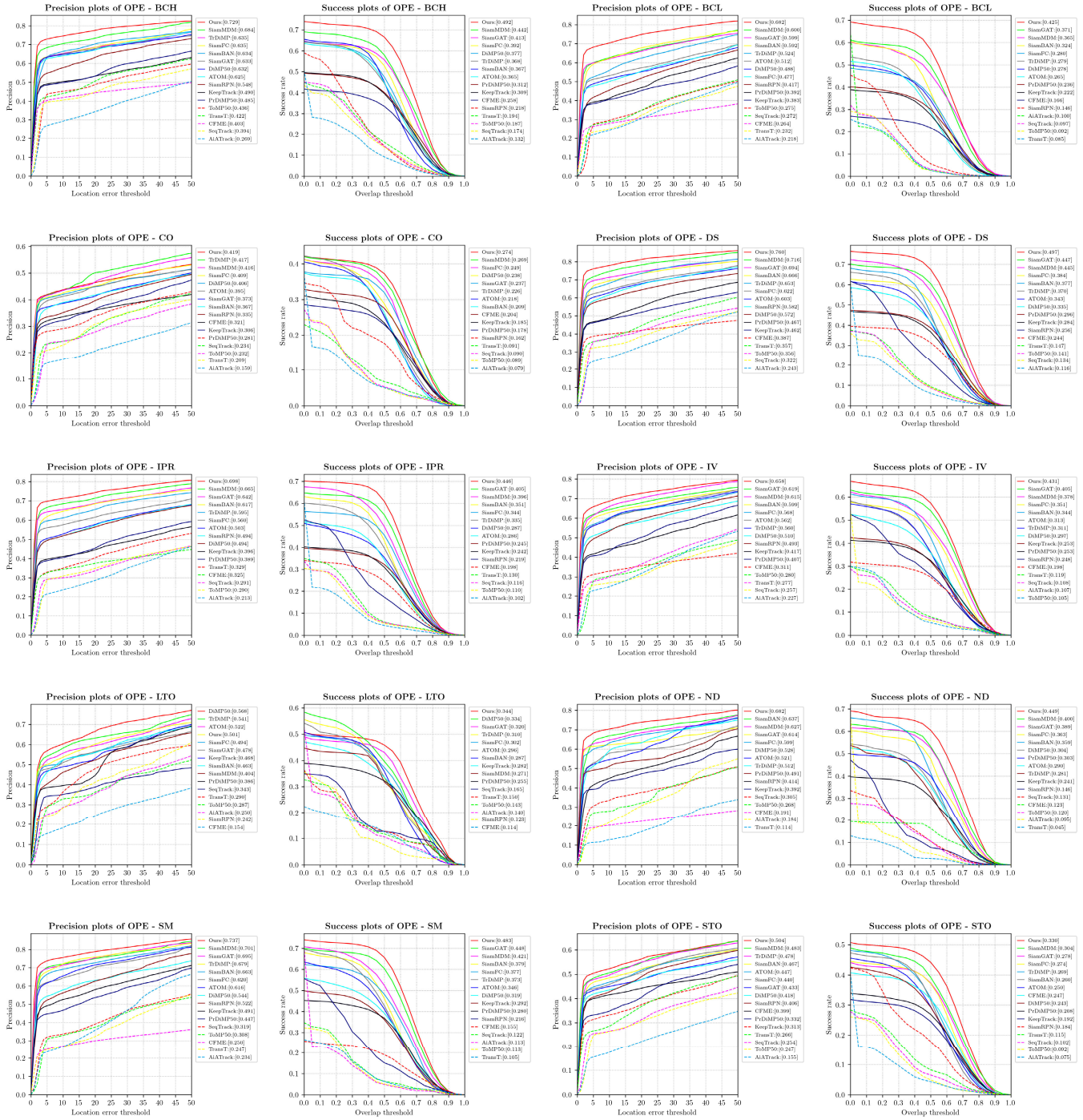



Fig. 8. Precision and success curves of each algorithm on different attributes of the SV248S dataset.

As shown in Fig. 8, it can be seen that the proposed algorithm performs the best in precision and success with the nine attributes of BCH, BCL, CO, DS, IPR, IV, ND, SM, and STO.

*3) Visual Analysis:* For a visual comparison, six video sequences with different challenging attributes were selected from the SV248S dataset for visual analysis, that is, seq1_13, seq2_02, seq3_10, seq5_18, seq5_21, and seq6_40. The visual results of SeqTrack, TransT, SiamMDM, ATOM, and the proposed SiamS$^2$F framework, in addition to the ground truth, are displayed in Fig. 9. The seq1_13 video sequence contains four challenging attributes: STO, DS, IV, and BCL. At frame 81, the SeqTrack, TransT, and ATOM algorithms fail to track the target because there are many similar objects around the tracking target. The proposed tracker can achieve correct tracking because the algorithm can remove some of the contextual interference based on the center loss function. When the tracking object is at frame 182, the SiamMDM algorithm fails to track because the target is occluded. The proposed SiamS$^2$F tracker can still follow the target, which indicates that considering the spectral, spatial, and frame information and focusing on the continuity of the target between frames is an effective approach for satellite video tracking. When the target is temporarily occluded, the proposed tracker can capture it again if the target appears in the adjacent frames. The seq2_02

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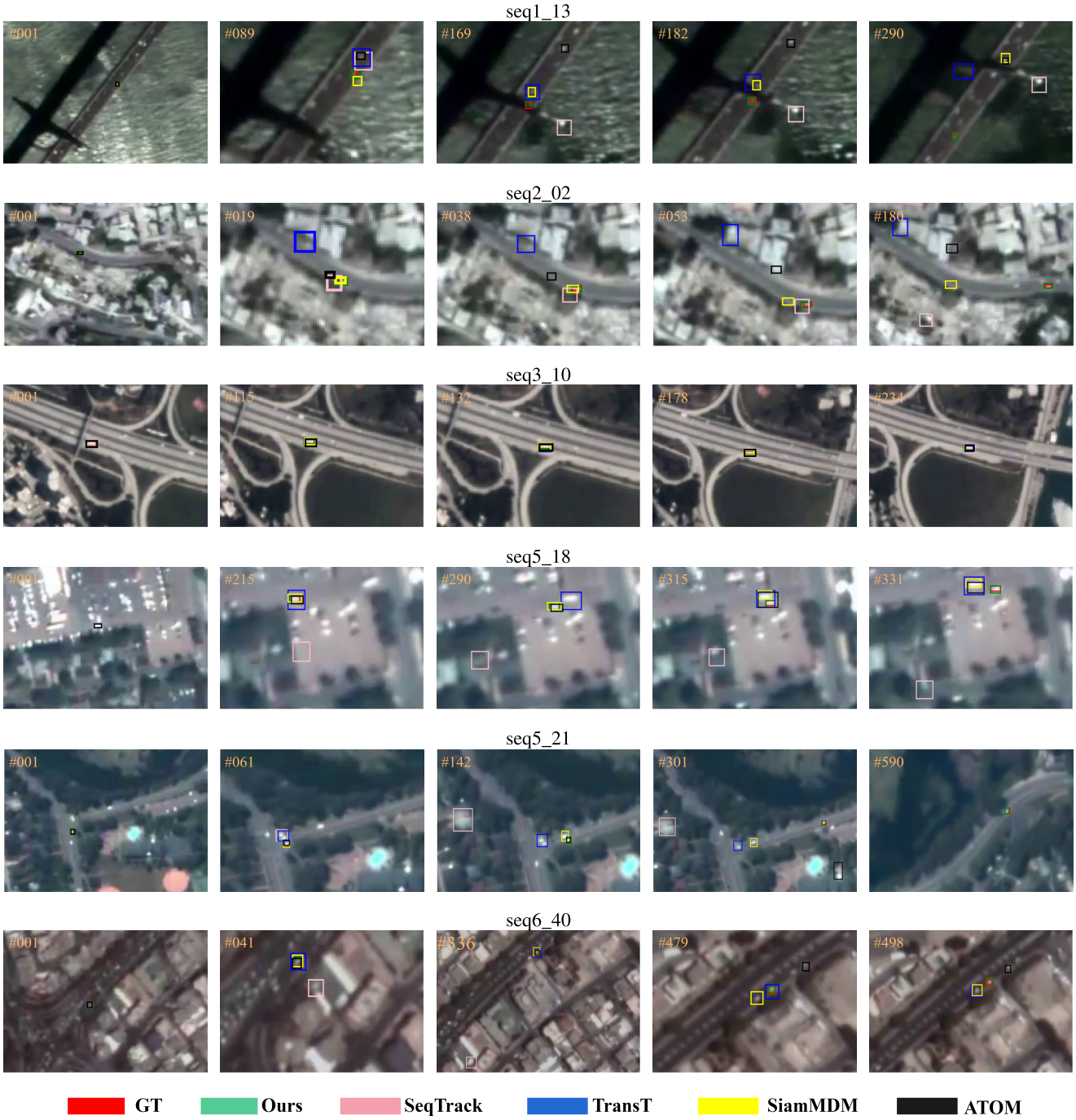



Fig. 9. Qualitative comparison of the state-of-the-art visual trackers on the seq1_13, seq2_02, seq3_10, seq5_18, seq5_21, and seq6_40 video sequences from the SV248S dataset. The frame number is shown on each frame.

video sequence contains four challenging attributes: DS, ND, BCL, and IPR. In this sequence, the tracked object in the video frame has less distinct shape and color characteristics, due to being affected by the complex background. This results in it being very challenging for the other methods to acquire accurate results. The proposed SiamS$^2$F tracker also works well on this kind of sequence because the proposed SSF module has a good pixel-level target recognition ability, and the local information of the target can be modeled to achieve a good tracking result. The seq3_10 video sequence contains two challenging attributes: DS and BCH. It can be seen that all of the algorithms achieve good tracking results because the target is clear and the similar objects are relatively far from the object of interest. In the seq5_18 and seq5_21 video sequences, there are also some similar objects around the real target, but the proposed tracker can still achieve good results, compared with the other methods. The seq6_40 video sequence contains two challenging attributes—STO and DS—and there are occlusion and dense object similarity challenges. SeqTrack loses the target at frame 336, and SiamMDM and ATOM lose the target at frame 479. The proposed SiamS$^2$F tracker tracks the true target successfully in the displayed frames, which shows the potential of the proposed method in the case of extremely complex scenes.

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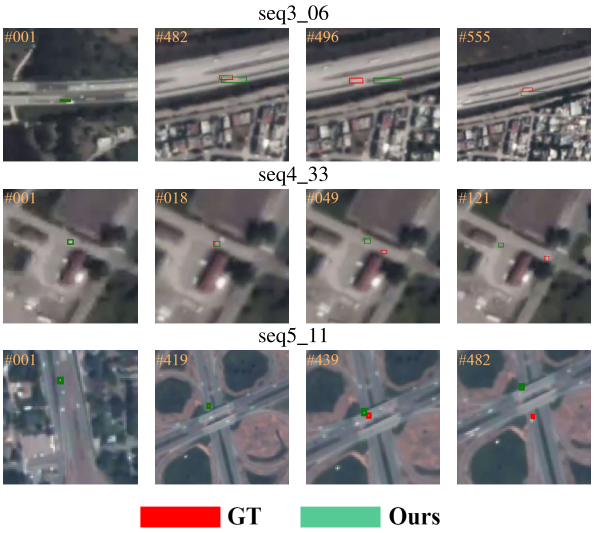



Fig. 10. Failure cases of the SiamS$^2$F network in three video sequences.

We also display some failure cases in Fig. 10. In the seq3_06 video sequence where most of the frames are blurry, it can be observed that the target in the initial frame is very small and there is no obvious difference between the color of the target vehicle and the color of the road surface. Under such a complex background, the proposed SiamS$^2$F tracker can still achieve a long accurate tracking period up to frame 482. After frame 482, SiamS$^2$F fails to capture the target information due to the high similarity between the target and the background information. In seq4_33, there are multiple challenges such as DS, IV, SM, and IPR, and the proposed tracker can successfully complete the tracking task in the first 20 frames. After this, the target vehicle cannot be tracked because a 90° direction change of the trajectory appears. This shows that the proposed SiamS$^2$F framework shows a good tracking performance for a target with a straight trajectory, but the potential of the network could be further improved to tackle scenes where the direction of the trajectory changes greatly. One of the most challenging attributes in the seq5_11 video sequence is CO (i.e., two or more full occlusions in the video), where the proposed tracker is unable to predict the location of the target due to a constant lack of motion information and the presence of many similar objects nearby.

## V. Conclusion

In this article, we have proposed a Siamese classification and regression framework called SiamS$^2$F for end-to-end satellite video object tracking. In the SiamS$^2$F framework, the initial deep features of the template branch and the search branch are extracted based on the Siamese network structure, along with the parallel CSM. The SSF feature fusion module, which contains the graph attention module and the MFIA module, is then introduced to better consider the interframe spectral and spatial information of the target under the situation of a target with few pixels. To better locate the real target among similar objects with a complex background, an additional center loss function is proposed, which reduces the impact of the similar objects by introducing a constraint that considers the center point of the search box and the real target. The results confirmed the potential of the proposed method in tracking moving targets in satellite videos under complex scenes with comprehensive interference factors, such as extremely small targets, similar objects, and occlusions.

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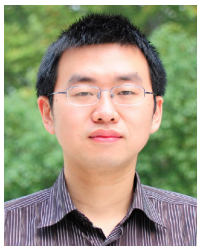
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